

13. $\int (\ln x)^2 dx$ (usando substituição)

$$\begin{aligned} u &= (\ln x)^2 & \Rightarrow & u' = 2(\ln x) \cdot \frac{1}{x} \\ v' &= 1 & v &= x \end{aligned}$$

$$\begin{aligned} \int (\ln x)^2 dx &= \int uv' = uv - \int u'v \\ &= (\ln x)^2 x - \int [2(\ln x) \cdot \frac{1}{x}] x dx \\ &= (\ln x)^2 x - 2 \int \ln x dx \end{aligned}$$

Note: $\int \ln x dx = \int uv' dx$ $u = \ln x$
 $v' = 1$

$$\begin{aligned} &= uv - \int u'v dx \\ &= (\ln x)x - \int \left(\frac{1}{x}\right)x dx \\ &= (\ln x)x - \int dx = (\ln x)x - x + C \end{aligned}$$

Então, $\int (\ln x)^2 dx = (\ln x)^2 x - 2x \ln x + 2x + C.$

33. $\int \sin \sqrt{x} dx$ (usando uma substituição e depois por partes).

$$y = \sqrt{x} \Rightarrow x = y^2 \Rightarrow dx = 2y dy$$

$$\Rightarrow \int \sin \sqrt{x} \, dx = \int \sin y (2y \, dy)$$

$$= 2 \int y \sin y \, dy$$

$$\begin{aligned} u &= y \\ v' &= \sin y \end{aligned} \Rightarrow \begin{aligned} u' &= 1 \\ v &= -\cos y \end{aligned}$$

$$= 2 \int u v'$$

$$= 2 \left(uv - \int u' v \right)$$

$$= 2 \left(-y \cos y - \int (-\cos y) \, dy \right)$$

$$= -2y \cos y + 2 \int \cos y \, dy$$

$$= -2y \cos y + 2 \sin y + C$$

$$= -2\sqrt{x} \cos \sqrt{x} + 2 \sin \sqrt{x} + C$$

35. $\int_{\frac{\sqrt{\pi}}{2}}^{\sqrt{\pi}} \theta^3 \cos(\theta^2) \, d\theta$ (usando uma substituição e depois por partes).

Fazendo $x = \theta^2$, temos

$$\theta = x^{1/2} \Rightarrow d\theta = \frac{1}{2} x^{-1/2} \, dx$$

$$\Rightarrow \int_{\frac{\sqrt{\pi}}{2}}^{\sqrt{\pi}} \theta^3 \cos(\theta^2) \, d\theta = \int_{\frac{\pi}{2}}^{\pi} (x^{1/2})^3 \cos x \left(\frac{1}{2} x^{-1/2} \, dx \right)$$

$$= \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} x \cos x \, dx$$

$$\begin{aligned} u &= x & \Rightarrow & u' = 1 \\ v' &= \cos x & v &= \sin x \end{aligned}$$

$$= \frac{1}{2} \int_{\pi/2}^{\pi} u v'$$

$$= \frac{1}{2} \left(u v \Big|_{\pi/2}^{\pi} - \int_{\pi/2}^{\pi} u' v \right)$$

$$= \frac{1}{2} \left(x \sin x \Big|_{\pi/2}^{\pi} - \int_{\pi/2}^{\pi} \sin x \, dx \right)$$

$$= \frac{1}{2} \left(-\frac{\pi}{2} + \cos x \Big|_{\pi/2}^{\pi} \right)$$

$$= \frac{1}{2} \left(-\frac{\pi}{2} - 1 \right)$$

Integrais trigonométricas.

$$18. \int \sin^3 x \sqrt{\cos x} \, dx$$

$$\begin{aligned} u &= \cos x \\ du &= -\sin x \, dx \end{aligned}$$

$$= \int (1 - \cos^2 x) \sqrt{\cos x} \cdot \sin x \, dx$$

$$= \int (1 - u^2) \sqrt{u} (-du)$$

$$= \int (u^{5/2} - u^{3/2}) \, du = \frac{2}{7} u^{7/2} - \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{7} \cos^{7/2} x - \frac{2}{3} \cos^{3/2} x + C$$

$$\begin{aligned}
 17. \quad \int \cos^2 x \tan^3 x \, dx &= \int \sin^3 x (\cos x)^{-1} \, dx \\
 &= \int (1 - \cos^2 x) (\cos x)^{-1} \sin x \, dx \quad u = \cos x \\
 &= \int (1 - u^2) u^{-1} (-du) \\
 &= \int (u - u^{-1}) \, du = \frac{u^2}{2} - \ln |u| + C \\
 &= \frac{\cos^2 x}{2} - \ln |\cos x| + C.
 \end{aligned}$$

$$41. \quad \int \sin(5x) \sin(2x) \, dx$$

$$= \int \frac{\cos(3x) - \cos(7x)}{2} \, dx$$

$$= \frac{\sin(3x)}{6} - \frac{\sin(7x)}{14} + C$$

$$\begin{aligned}
 \sin(a) \sin(b) \\
 &= \frac{\cos(a-b) - \cos(a+b)}{2}
 \end{aligned}$$

Substituição trigonométrica

$$5. \quad \int_0^2 \frac{dt}{\sqrt{2} \, t^3 \sqrt{t^2 - 1}}$$

$$\begin{aligned}
 t &= \sec \theta \\
 \Rightarrow t^2 - 1 &= \sec^2 \theta - 1 \\
 &= \frac{1}{\cos^2 \theta} - 1 = \tan^2 \theta
 \end{aligned}$$

$$\begin{aligned}
 \frac{dt}{d\theta} &= \frac{d}{d\theta} \left(\frac{1}{\cos \theta} \right) = \frac{-1}{\cos^2 \theta} \cdot (-\sin \theta) \\
 &= \sin \theta / \cos^2 \theta.
 \end{aligned}$$

$$\int_{\sqrt{2}}^2 \frac{dt}{t^3 \sqrt{t^2-1}} = \int_{\operatorname{arcsec}(\sqrt{2})}^{\operatorname{arcsec}(2)} \frac{(\sin \theta / \cos^2 \theta) d\theta}{\sec^3 \theta |\tan \theta|}$$

Note: $a = \operatorname{arcsec}(\sqrt{2})$

$$\Leftrightarrow \sec(a) = \sqrt{2}$$

$$\Leftrightarrow \cos(a) = 1/\sqrt{2} = \sqrt{2}/2$$

$$\Rightarrow a = \frac{\pi}{4}$$

$$b = \operatorname{arcsec}(2)$$

$$\Leftrightarrow \sec(b) = 2 \Leftrightarrow \cos(b) = 1/2 \Leftrightarrow b = \frac{\pi}{3}$$

Logo,

$$\int_{\sqrt{2}}^2 \frac{dt}{t^3 \sqrt{t^2-1}} = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{(\sin \theta / \cos^2 \theta) d\theta}{\sec^3 \theta |\tan \theta|}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cancel{\sin \theta} \cdot (\cos \theta)^{-2}}{(\cos \theta)^{-3} \cancel{\sin \theta} (\cos \theta)^{-1}} d\theta$$

$\tan \theta > 0$
em $[\frac{\pi}{4}, \frac{\pi}{3}]$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos^2 \theta d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1 + \cos(2\theta)}{2} d\theta$$

$$= \left(\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right) \Big|_{\pi/4}^{\pi/3}$$

$$= \frac{\pi}{6} + \frac{\sin(2\pi/3)}{4} - \frac{\pi}{8} - \frac{\sin(\pi/2)}{4}$$

$$= \frac{\pi}{24} + \frac{\sqrt{3}}{8} - \frac{1}{4}$$

Frações Parciais

8. $\int \frac{r^2}{r+4} dr$

Note:

$$\begin{array}{r} r^2 \overline{) r+4} \\ -(r^2+4r) \\ \hline -4r \\ -(-4r-16) \\ \hline 16 \end{array}$$

$$\therefore r^2 = (r+4)(r-4) + 16$$

$$\Rightarrow \int \frac{r^2}{r+4} dr = \int \left(r-4 + \frac{16}{r+4} \right) dr$$

$$= \frac{r^2}{2} - 4r + 16 \cdot \ln |r+4| + C$$

21. $\int \frac{5x^2+3x-2}{x^3+2x^2} dx$

Note: $x^3+2x^2 = x^2(x+2)$

Escrevamos

$$\frac{5x^2+3x-2}{x^3+2x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$$

$$= \frac{Ax+B}{x^2} + \frac{C}{x+2}$$

$$= \frac{(Ax+B)(x+2) + cx^2}{x^2(x+2)}$$

Queremos determinar A, B e C tais que

$$5x^2 + 3x - 2 = (Ax+B)(x+2) + cx^2$$

$$= Ax^2 + 2Ax + Bx + 2B + cx^2$$

$$= (A+C)x^2 + (2A+B)x + 2B$$

$$\therefore A+C = 5$$

$$\Rightarrow B = -1$$

$$2A+B = 3$$

$$\Rightarrow 2A - 1 = 3 \Rightarrow A = 2$$

$$2B = -2$$

$$\Rightarrow C = 5 - A = 3$$

$$\therefore \int \frac{5x^2 + 3x - 2}{x^3 + 2x^2} dx = \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3}{x+2} \right) dx$$

$$= 2 \ln|x| + \frac{1}{x} + 3 \ln|x+2| + K$$

Substituição racionalizante

$$39. \int \frac{dx}{x\sqrt{x+1}}$$

$$u = \sqrt{x+1}$$

$$\Rightarrow u^2 = x+1 \Rightarrow x = u^2 - 1$$

$$\Rightarrow dx = 2u du$$

$$\Rightarrow \int \frac{dx}{x\sqrt{x+1}} = \int \frac{2u du}{(u^2-1)u} = 2 \int \frac{du}{u^2-1}$$

Note: $u^2-1 = (u+1)(u-1)$.

Escrevamos:

$$\frac{1}{u^2-1} = \frac{A}{u+1} + \frac{B}{u-1}$$

$$= \frac{A(u-1) + B(u+1)}{u^2-1}$$

$$\Rightarrow 1 = A(u-1) + B(u+1) = (A+B)u - A+B$$

$$\Rightarrow \begin{array}{l} A+B=0 \\ -A+B=1 \end{array} \Rightarrow \begin{array}{l} A = -1/2 \\ B = 1/2 \end{array}$$

$$\therefore \int \frac{dx}{x\sqrt{x+1}} = 2 \int \left(\frac{A}{u+1} + \frac{B}{u-1} \right) du$$

$$= \int \left(\frac{-1}{u+1} + \frac{1}{u-1} \right) du = -\ln|u+1| + \ln|u-1| + K$$

$$= -\ln|\sqrt{x+1}+1| + \ln|\sqrt{x+1}-1| + K$$